

TrES-3b

R-0416

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230801 · created 2026-07-19T07:07:55+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460157.7918 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.017
Transit depth (Rp/Rs)^2	3.78 +/- 0.67 [%]
Orbital Inclination (inc)	82.33 +/- 0.87
Airmass coefficient 1 (a1)	1.036 +/- 0.015
Airmass coefficient 2 (a2)	-0.027 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.45 %
Best Comparison Star	#1 - [340, 359]
Optimal Method	PSF photometry
Transit Duration (day)	0.0618 +/- 0.0073

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.194 ± 0.017 vs 0.16309 (deviation 1.74σ)
Duration fit vs published	0.0618 d vs 0.0567 d (deviation 0.67σ)
Mid-time O-C	-5.2 min
Depth SNR	10.9
Residual scatter	1.45 %

Light curve

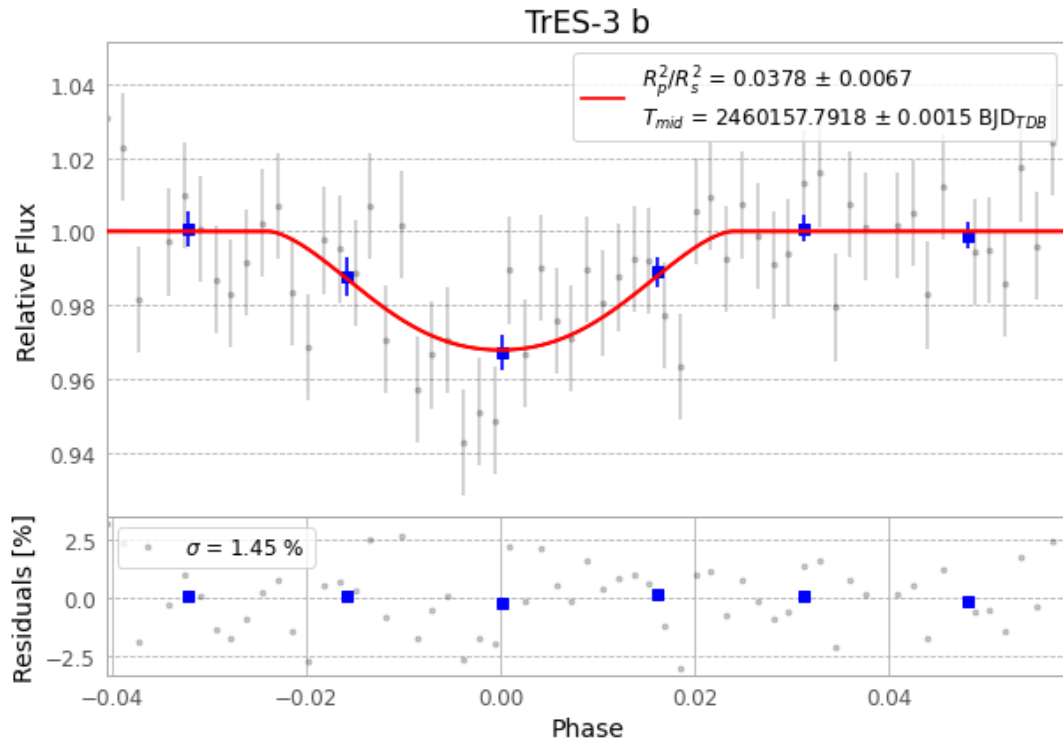


Figure 1 — FinalLightCurve_TrES-3 b_2023-07-31.png (SHA-256 6928d458b4f60298...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [231, 202], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 294c71cad6c69966...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

60 FITS frames (39 MB), TRES-3230801053916.FITS → TRES-3230801084515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230801.log (5c15a5701d8a...), FinalLightCurve_TrES-3 b_2023-07-31.png (6928d458b4f6...), FinalLightCurve_TrES-3 b_2023-07-31.csv (8adbc334724c...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 07:07Z.